**試題:**

- Please give details of your calculation. A direct answer without explanation is not counted.
- Your answers must be in English.

# Problem 1 (15%)

What is the bit string of $-37$ under IEEE 754 standard? We assume single precision. Please explain details instead of just giving a string.

# Problem 2 (30%)

Consider the following matrix:

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 10 \end{bmatrix}$$

We would like to find the LU factorization using the pivoting.

(a) Give all details of finding

$$M_{n-1}P_{n-1}\dots M_1P_1A = U$$

(b) Find $P$ such that

$$PA = LU \tag{1}$$

where $L$ and $U$ are lower- and upper-triangular matrices, respectively.

(c) If we solve the linear system

$$Ax = b = \begin{bmatrix} 6 \\ 9 \\ 12 \end{bmatrix}$$

show how to use (1) to solve the linear system.

# Problem 3 (20%)

Consider the following matrix:

$$A = \begin{bmatrix} 16 & 8 & 4 & 12 \\ 8 & 8 & 0 & 2 \\ 4 & 0 & 27 & 0 \\ 12 & 2 & 0 & 30 \end{bmatrix}$$

Generate the Cholesky factorization using the outer product form.

# Problem 4 (20%)

In exact arithmetic, the addition operation is commutative, i.e.,

$$x + y = y + x$$

for any two numbers $x, y$ and also associative, i.e.,

$$x + (y + z) = (x + y) + z$$

for any $x, y$ and $z$. Is the floating point addition operation commutative? Is it associative? We assume exact rounded operations.

# Problem 5 (15%)

What is the smallest positive denormalized number under the IEEE standard (single precision)?